Bangladesh Army University of Engineering & Technology (BAUET)

Qadirabad Cantonment, Natore-6431

Faculty of Electrical and Computer Engineering

Department of Computer Science and Engineering

Lecture Plan

Program:	**Computer Science and Engineering (CSE)**
Course Code & Title:	MATH-2141 (Vector Analysis, Matrices and Laplace Transformation)
Semester:	Summer-2025
Level:	2nd Year 1st Semester
Credit Hour:	3.00 Hrs./Week
Name & Designation of Teacher:	Md. Abdullah Al Mamun Assistant Professor & Head, Dept. of Mathematics
Class Hours:	Monday: 10:15-11:10
	Tuesday:09:15-10:10
	Wednesday: 09:15-10:10
Consultation Hours:	
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Rationale:	In this course, students will learn at the basic properties of vectors and scalars, gradient, divergence and curl, line integration, surface integration, volume integration, Matrices as objects rather than just as a tool for solving systems of linear equations, Laplace transform is used to solve the ordinary and partial differential equation.
Pre-requisite (if any):	Differential and Integral Calculus

Course Objective: The course objectives include the following:	
1.	To analyze the characteristics of scalar and vector valued functions.
2.	To determine relationship between coefficient matrix invertibility and solutions to a system of linear equations.
3.	To solve ordinary and partial differential equation with the help of Laplace Transform.

Course Outcomes (CO):

Upon completion of the course, the students will be able to:

CO1	Understand about product of vectors, gradient, divergence, curl, matrix, rank, eigen-values, and Laplace transform.
CO2	Able to solve the problem of vector, vector differentiation, vector integration, rank nullity, and Laplace transform.
CO3	Apply the knowledge of gradient, divergence, curl, matrix, and Laplace transform in practical field.

Course Content:

Vector Analysis: Scalars and vectors, equality of vectors, addition and subtraction of vectors, multiplication of vectors by scalars, scalar and vector product of two vectors and their geometrical interpretation, triple products and multiple products, linear dependence and independence of vectors, differentiation and integration of vectors along with elementary applications, vector geometry, gradient, divergence and curl of point functions, definition of line, surface and volume integrals, Green's theorem, divergence theorem of Gauss's, Stoke's theorem and their applications.
Matrices: Definition of matrix, algebra of matrices, transpose of a matrix, inverse of matrix, rank and elementary transformations of matrices, solution of linear equations, linear dependence and independence of vectors, quadratic forms, matrix polynomials, determination of characteristic root and vectors, Cayley-Hamilton theorem and it's application, null space and nullity of matrix, characteristic subspace of matrix.
Laplace Transform: Definition, Laplace transformations of some elementary functions, sufficient conditions for existence of Laplace transformation, some important properties of Laplace transformations, Laplace transformations of derivatives, the unit step function, periodic functions, some special theorems on Laplace transformations, partial fraction, inverse Laplace transformations & their properties and applications of Laplace transformations to solve boundary value problems.

Student's Responsibilities:

- Students must come to the class prepared for the course material covered in the previous class
- They must submit their assignments on time
- No late or partial assignments will be acceptable
- There will be no make-up Class Test

Grading Policy:

Numerical Score	Letter Grade		Grade points
80% and above	A+	A (Plus)	4.00
75% to > 80%	A	A (Regular)	3.75
70% to > 75%	A-	A (Minus)	3.50
65% to > 70%	B+	B (Plus)	3.25
60% to > 65%	B	B (Regular)	3.00
55% to > 60%	B-	B (Minus)	2.75
50% to > 55%	C+	C (Plus)	2.50
45% to > 50%	C	C (Regular)	2.25

40% to > 45%	D	D (Pass)	2.00
below 40%	F	Fail	0.00
Incomplete	I	-	-
Withdrawal	W	-	-
Project / Thesis Continuation	X	-	-

Generic Skills: Graduate Attributes/Program Outcomes (POs)

Program Outcomes (POs) represent the knowledge, skills and attitudes the students should have at the end of a four years engineering program. CSE program of BAUET has 12 Program Outcomes. They are briefly described in the following table.

Sl.No	PO	Category	Description
1	**PO 1**	**Engineering Knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2	**PO 2**	**Problem Analysis**	Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	**PO 3**	**Design/Development of Solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, social and environmental concerns.
4	**PO 4**	**Investigation**	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
5	**PO 5**	**Modern Tool Usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	**PO 6**	**The Engineer and Society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	**PO 7**	**Environment and Sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of need for sustainable development.
8	**PO 8**	**Ethics**	Apply ethical principles and commit to professional Ethics and responsibilities and norms of the engineering practice.
9	**PO 9**	**Individual and Team Work**	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	**PO 10**	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	**PO 11**	**Project Management and Finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	**PO 12**	**Life Long Learning**	Recognize the need for, and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

POs	Graduate Attributes (Keywords)	WA-WK's	WP/EA
1	Engineering Knowledge	WK1-WK4	WP
2	Problem Analysis	WK1-WK4	WP
3	Design/Development of Solutions	WK5	WP
4	Investigation	WK8	WP
5	Modern Tool Usage	WK6	WP
6	The Engineer and Society	WK7	WP
7	Environment and Sustainability	WK7	WP
8	Ethics	WK7	--
9	Individual and Team work	--	--
10	Communication		EA
11	Individual and Team Work	---	--
12	Project Management and Finance	--	--
Assessments provide adequate feedback to the program to identify strengths and weaknesses for CQI			

Complex Engineering Problems (Need High Taxonomy Level)

Complex Engineering Problems have characteristic WP1 and some or all of WP2 to WP7, EP1 and EP2, that can be resolved with in-depth forefront knowledge

WP1	Depth of Knowledge required	Resolved with **forefront in-depth** engineering knowledge (WK3, WK4, WK5, WK6 or WK8) which allows a fundamentals-based, first principles analytical approach
WP2	Range of conflicting requirements	Involve **wide-ranging or conflicting** technical, engineering and other issues.
WP3	Depth of analysis required	Have **no obvious solution** and require abstract thinking, originality in analysis to formulate suitable models.
WP4	Familiarity of issues	Involve **infrequently encountered** issues
WP5	Extent of applicable codes	**Beyond codes** of practice
WP6	Extent of stakeholder involvement and level of conflicting requirements	Involve **diverse groups of stakeholders** with widely **varying needs**.
WP7	Interdependence	Are **high level problems** including **many component** parts or sub-problems.
EP1	Consequences	Have **significant consequences** in a range of contexts.
EP2	Judgement	Require judgement in **decision** making

Complex Engineering Activities (Project based)

Activities	Preamble	**Complex activities** means (engineering) activities or projects that have **some or all** of the following characteristics listed below
EA1	Range of resources	**Diverse resources** (people, money, equipment, materials, information and technologies).
EA2	Level of interaction	Require resolution of significant problems arising from interactions between **wide ranging** or **conflicting** technical, engineering or other issues.
EA3	Innovation	Involve creative use of engineering principles and **research- based** knowledge in **novel** ways.
EA4	Consequences to society and the environment	Have **significant consequences** in a **range of contexts**, characterised by **difficulty** of prediction and mitigation.
EA5	Familiarity	Can extend **beyond previous** experiences by applying **principles-based** approaches.

Bloom's Taxonomy Domain:

Cognitive Domain		**Psychomotor Domain**		**Affective Domain**	
C1	Remembering	**P1**	Perception	**A1**	Receive
C2	Understanding	**P2**	Set	**A2**	Respond
C3	Applying	**P3**	Guided Response	**A3**	Value
C4	Analyzing	**P4**	Mechanism	**A4**	Organize
C5	Evaluating	**P5**	Complex Overt Response	**A5**	Internalize
C6	Creating/Designing	**P6**	Adaption		
		P7	Origination		

Detailed Mapping (According to BAETE Accreditation Manual- II)

Course Code & Title: MATH-2141 (Vector Analysis, Matrices and Laplace Transform**)**

Mapping CO-PO, Bloom's Level, Knowledge Profile, Complex Engineering Problem (WP) and Complex Engineering Activities (EA)

Course Outcomes (CO)	Statements	PO/ WA	PI Code	Bloom's Taxonomy Level	Delivery methods and activities	Assessment Tools
CO1	Understand about product of vectors, gradient, divergence, curl, matrix, rank, eigen-values, and Laplace transform.	1	1.1.1	C1, C2	Lectures, Text Book, PPT, PDF White Board Writing	Class Participation / Observation Homework's, Assignments, Class Tests, Semester Final
CO2	Able to solve the problem of vector, vector differentiation, vector integration, rank nullity, and Laplace transform.	2	1.1.1	C2		
CO3	Apply the knowledge of gradient, divergence, curl, matrix, and Laplace transform in practical field.	2	1.1.1	C2,C3		

						Examination (Written)

Legends: CO-Course Outcomes PO-Program Outcomes, KP-Knowledge Profile
Bloom's Level: Cognitive Domain (C1-C6), Psychomotor Domain (P1-P7), and Affective Domain (A1-A5)

Course Outcomes (CO)	Program Outcome (PO)											
	1	2	3	4	5	6	7	8	9	10	11	12
CO1	H											
CO2		H										
CO3		H										

Legends: High (H)=3, Medium(M)=2 and Low(L)=1

Assessment of CO-PO, Bloom's Level

Teaching-learning and Assessment Strategy:
Class Participation/Presentation, Class Performance, Homework/Assignment, Class Test, Semester Final Examination (Written)

Assessment Methods	Description	Grading	CO	Blooms Taxonomy	
Continuous Assessment (40%)	Class Test/ Homework/Assignment	15%	CO1 CO2 CO3	C1, C2, C3	Question 1 and 5 will satisfy specified CO- POs, BLs, KPs and WPs in the semester final examination
	Class Participation/Presentation Skill	5%	CO1 CO2 CO3	C1, C2, C3	
	Mid-Term Examination	20%	CO3	C1, C2, C3	
Final Examination (60%)		60%	CO1 CO2 CO3	C1, C2, C3	
Total Marks		100%			

Teaching and Learning Strategies (TLS)	
TLS1	Delivery lectures using multimedia of different topics
TLS2	Active and fruitful discussion in class regarding efficient solving of the logical, mathematical and complex engineering problems
TLS3	Group discussion and presentation regarding diverse problems and corresponding lectures
TLS4	Evaluation of class performance to reach each student in a class for every topic.

Course Outline

S.N	Lectures	Topics	Remark
1.	Lecture: 1	Introduction, Scalars, and vectors, equality of vectors, unit vector, examples.	
2.	Lecture: 2	Addition and subtraction of vectors, Multiplication of vectors by scalars, examples	
3.	Lecture: 3	Linear dependence and independence of vectors, moments, example	
4.	Lecture: 4	Scalar product of two vectors, their geometrical interpretation, and problem-solving	
5.	Lecture: 5	Vector product of two vectors, their geometrical interpretation, and problem-solving	
6.	Lecture: 6	Triple product, multiple products, and problem-solving	
7.	Lecture: 7	Area, volume, coplanar of vector, examples.	
8.	Lecture: 8	Review class and problem-solving	
9.	Lecture: 9	Moment, the physical meaning of dot and cross product	
10.	Lecture:10	Differentiation of vectors, components of velocity and acceleration, problem-solving	
11.	Lecture: 11	Partial differentiation of vectors and problem solving	
12.	Lecture: 12	Gradient, definition, physical meaning, problem-solving	
13.	Lecture: 13	Divergence, definition, physical meaning, problem-solving	
14.	Lecture: 14	Curl, definition, physical meaning, problem-solving	
15.	Lecture: 15	Solenoidal of vector, irrotational properties of vector, problem-solving.	
16.	Lecture: 16	Vector integration, line integration, problem-solving	
17.	Lecture: 17	Surface integration, problem-solving	
18.	Lecture: 18	Volume integration, problem-solving	
19.	Lecture: 19	Green's Theorem, statement, proof, problem-solving	
20.	Lecture: 20	Divergence Theorem, Stokes Theorem, statement, problem-solving	
21.	Lecture: 21	Class test from previous classes	
		Mid-Break	
22.	Lecture: 22	Introduction, matrix, definition, types of matrices, examples. Adjoint matrix, the inverse of a matrix	
23.	Lecture: 23	Matrix polynomials, eigen-values	
24.	Lecture: 24	Eigenvectors, problem solving	
25.	Lecture: 25	Cayley-Hamilton theorem with example and application in various fields.	
26.	Lecture: 26	Rank and nullity	
27.	Lecture: 27	Normal and canonical forms of matrix	
28.	Lecture: 28	Definition, Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace Transforms	
29.	Lecture: 29	Some important properties of Laplace transform	
30.	Lecture: 30	Laplace Transforms of first translation and second translation properties, Laplace Transforms of derivatives	
31.	Lecture: 31	The unit step function, Periodic function	
32.	Lecture: 32	Problem solving from periodic function	
33.	Lecture: 33	Inverse Laplace Transforms	
34.	Lecture: 34	Properties of Inverse Laplace Transforms	
35.	Lecture: 35	Problem solving from Inverse Laplace Transforms	
36.	Lecture: 36	Convolution theorem	
37.	Lecture: 37	Problem solved by using convolution theorem	
38.	Lecture: 38	Partial fractions	
39.	Lecture: 39	Problem solved by using partial fractions	
40.	Lecture: 40	Solutions of differential equations by Laplace Transforms	
41.	Lecture: 41	ODE and PDE by Laplace Transforms	
42.	Lecture: 42	Review class	

Text and Ref Books:	
1.	Vector Analysis - Seymour Lipschutz, Dennis Spellman and Murray R. Spiegel, 2nd Edition, ISBN-13: 978-0071615457
2.	Theory and Problems of Matrices, McGraw-Hill-Frank Ayres, Jr, 1st Edition, ISBN-13: 978-0070026568
3.	Matrices- M.L.Khanna
4.	Mathematical Methods- Prof. AbdurRahman
5.	Laplace Transform by Murry R Spiegel, Phd

Detail Lecture Plan

Week & Class No.		Topic to be covered	CO	Teaching & Learning Strategies (TLS)
Week-1: Introduction (Vector and Scalar) **Learning Outcomes:** Able to identify the fundamental properties of scalar, vector, unit vector, algebra of vector, resultant vector, linearly dependence, and linearly independence				
W-1	C-1	Introduction, Scalars, and vectors, equality of vectors, unit vector, examples.	CO1	**TLS1**
	C-2	Addition and subtraction of vectors, Multiplication of vectors by scalars, examples	CO1	
	C-3	Linear dependence and independence of vectors, moments, example	CO2	
Reference: Vector Analysis, Murray R. Spiegel, Ph. D, Chapter 1, Vector and Scalar Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors				
Applications:				
• A very good application of vectors is Global Positioning System (GPS). • A good application for vector analysis is the calculation of an airplane's flight path. This requires knowing the magnitude and direction of both the plane's thrust, as well as that of the prevailing winds acting on the plane. • Navigating by air and by boat is generally done using vectors. • Planes are given vectors to travel and they use their speed to determine how far they need to go before turning. Flight plans are made using a series of vectors.				
Review Question:				
1. Find the magnitude of $(\boldsymbol{A}+\boldsymbol{B})$ where $\boldsymbol{A}=12\boldsymbol{i}+6\boldsymbol{j}-9\boldsymbol{k}$ and $=-6\boldsymbol{i}+5\boldsymbol{j}+7\boldsymbol{k}$. 2. Find the absolute of A where $\boldsymbol{A}=-7\boldsymbol{i}+3\boldsymbol{j}-11\boldsymbol{k}$. 3. Prove that the vectors $\boldsymbol{A}=3\boldsymbol{i}+\boldsymbol{j}-2\boldsymbol{k}$, $\boldsymbol{B}=-\boldsymbol{i}+3\boldsymbol{j}+4\boldsymbol{k}$, $\boldsymbol{C}=4\boldsymbol{i}-2\boldsymbol{j}-6\boldsymbol{k}$ can form the sides of a triangle. 4. Explain vector and scalar with examples. The position vectors of a points P and Q are given by $\boldsymbol{r_1}=2\boldsymbol{i}+3\boldsymbol{j}-\boldsymbol{k}$, $\boldsymbol{r_2}=4\boldsymbol{i}-3\boldsymbol{j}+2\boldsymbol{k}$. Determine **PQ** in terms of **i**, **j**, **k,** and find its magnitude.				
Assignment:				
Topic:				

Week-2: Dot and Cross Product

Learning Outcomes:

Able to identify the fundamental of the dot product, cross product, and triple product

W-2	C-4	Scalar product of two vectors, their geometrical interpretation, and problem-solving	CO1	TLS2
	C-5	Vector product of two vectors, their geometrical interpretation, and problem-solving	CO2	TLS4
	C-6	Triple product, multiple products, and problem-solving	CO2	

Reference:

Vector Analysis, Murray R. Spiegel, Ph. D, Chapter 2, The dot and cross product

Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors

Applications:

- **Vector Product (Cross Product):** Cross product can be used in computer vision for tasks such as calculating the normal vector of a plane defined by three points, aiding in object recognition and scene understanding.
- **Scalar Product (Dot Product):** Dot product is used in computer vision algorithms for tasks like image processing, feature extraction, and pattern recognition.
- **Vector Triple Product:** The vector triple product is used in physics simulations for complex calculations involving angular momentum and torque.

Review Question:

1. Find the projection of the vector $\boldsymbol{A} = \boldsymbol{i} - 2\boldsymbol{j} + \boldsymbol{k}$ on the vector $\boldsymbol{B} = 4\boldsymbol{i} - 4\boldsymbol{j} + 7\boldsymbol{k}$.
2. Find the projection of the vector $4\boldsymbol{i} - \boldsymbol{3j} + \boldsymbol{k}$ on the line passing through the points (2, 3, -1) and (-2, -4, 3)
3. A wagon is pulled horizontally by exerting a constant force of 10 lb on the handle at an angle of 60^0 with the horizontal. How much work is done in moving the wagon 50 ft?
4. Find the area of a parallelogram having diagonals $\boldsymbol{A} = 3\boldsymbol{i} + \boldsymbol{j} - 2\boldsymbol{k}$ and $\boldsymbol{B} = \boldsymbol{i} - 3\boldsymbol{j} + 4\boldsymbol{k}$.

Assignment:

Topic:

Week-3: Dot and Cross Product

Learning Outcomes:

- Able to apply the knowledge of the basics of how to calculate work done

W-3	C-7	Area, volume, coplanar of vector, examples.	CO1	TLS2
	C-8	Review class and problem-solving	CO2	TLS4
	C-9	Moment, the physical meaning of dot and cross product	CO2	

Reference:
Vector Analysis by Murray R. Spiegel, Ph. D, Chapter 2, The dot and cross product
Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors

Applications:

- **Scalar Projection:** The dot product represents the scalar projection of one vector onto another. It quantifies how much one vector extends in the direction of another.
- **Work and Energy:** In physics, the dot product is used in the work-energy theorem. It represents the work done by a force along a particular direction.
- **Machine Learning:** In machine learning, the dot product is used in algorithms like support vector machines (SVM) for measuring similarity and in neural networks for weight updates.
- **Robotics**: In robotics, the cross product is used to compute rotational motion and velocity in robotic arms.

Review Question:

1. The angular velocity of a rotating body about an axis of rotation is given by $\boldsymbol{\omega} = 4\boldsymbol{i} + \boldsymbol{j} - 2\boldsymbol{k}$. Find the linear velocity of a point P on the body whose position vector relative to a point on the axis of rotation is $2\boldsymbol{i} - 3\boldsymbol{j} - \boldsymbol{k}$.
2. Find the work done in moving an object a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by $\boldsymbol{F} = 4\boldsymbol{i} - 3\boldsymbol{j} + 2\boldsymbol{k}$

Assignment:

Topic:

Class Test-1

Week & Class No.		Topic to be covered	CO	Teaching & Learning Strategies (TLS)
Week-4: Vector Differentiation **Learning Outcomes:** Able to identify and make an evaluation of the vector differentiation, partial differentiation, gradient, physical meaning of gradient				
W-4	C-10	Differentiation of vectors, components of velocity and acceleration, problem-solving	CO1	
	C-11	Partial differentiation of vectors and problem solving	CO2	
	C-12	Gradient, definition, physical meaning, problem-solving	CO2	
Reference: Vector Analysis by Murray R. Spiegel, Ph. D Chapter 3, Vector Differentiation Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors				
Applications:				

- **Vector Fields:** Vector differentiation is used to describe and manipulate vector fields, which are essential in computer graphics for modeling physical phenomena like fluid flow, electromagnetic fields, and deformable materials.
- **Gradient Descent:** Vector differentiation is a fundamental tool in machine learning for training models. It is used in gradient-based optimization algorithms to update model parameters and minimize the loss function.
- **Motion and Dynamics:** Vector differentiation is applied to describe the motion and dynamics of objects in physics simulations. It helps model and predict the behavior of particles, rigid bodies, and fluids in virtual environments.

Review Question:

1. A particle moves along the curve $x = 2t^2,\ y = t^2 + 4t,$ z= $3t - 5$, where t is the time. Find the components of its velocity and acceleration at time $t = 1$ in the direction $\boldsymbol{i} - 3\boldsymbol{j} + 2\boldsymbol{k}$
2. Find the unit tangent vector to any point on the curve $x = t^2 + 1$, y= $4t - 3$, $z = 2t^2 - 6t$. Determine the unit tangent at the point t= 3.
3. If $\boldsymbol{A} = (2x^2y - x^4)\boldsymbol{i} + (e^{xy} - ysinx)\boldsymbol{j} + (x^2cosy)\boldsymbol{k}$, find $\frac{\partial \boldsymbol{A}}{\partial x}, \frac{\partial \boldsymbol{A}}{\partial y}, \frac{\partial^2 \boldsymbol{A}}{\partial x^2}, \frac{\partial^2 \boldsymbol{A}}{\partial y^2}$.
4. Find the velocity and acceleration of a particle which moves along the curve $x = 2sin3t,\ y = 2cos3t,\ z = 8t$ at any time $t > 0$. Find the magnitude of the velocity and acceleration.
5. Show that $\boldsymbol{r} = e^{-t}(C\ cos2t + D\ sin2t)$, where C and D are constant vectors, is a solution of the differential equation $\frac{d^2\boldsymbol{r}}{d\boldsymbol{r}^2} + 2\frac{d\boldsymbol{r}}{dt} + 5\boldsymbol{r} = 0$.

Assignment:

Topic:

Week-5: Gradient, Divergence and Curl

Learning Outcomes:

Able to apply the knowledge of forwarding problems and can identify and evaluate the gradient, divergence, curl, physical meaning of gradient, divergence, curl

W-5	C-13	Divergence, definition, physical meaning, problem-solving	CO1	TLS2
	C-14	Curl, definition, physical meaning, problem-solving	CO2	TLS4
	C-15	Solenoidal of vector, irrotational properties of vector, problem-solving.	CO3	

Reference:

Vector Analysis by Murray R. Spiegel, Ph. D, Chapter 4, Gradient, Divergence, and Curl
Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors

Applications:

- **Divergence:** The easiest way to think about this is water coming out of a hose, the water will have a magnitude and a direction. This is essentially what divergence is. It can tell you how fast and in what direction fluid is moving. At the center of the toilet bowl, we have negative divergence because water is flowing inwards. At the rim, we have positive divergence as the water is moving away from the rim.
- **Gradient:** This is probably the closest thing to a regular derivative that you'll notice. A gradient essentially tells you how much a surface or some quantity changes from one point in space/time to another. Think of moving on a bed sheet, in different areas it will be bumpier than others, and how much it changes if the gradient.
- **Curl:** This is the easiest one of the three to understand conceptually, and that is because the curl tells you how much something twists or "Curls". You can imagine a toilet bowl flushing, if you wanted to know how much the water twisted in a certain section you would do that using the curl.

Review Question:

1. Show that $\boldsymbol{A} = (2x^2 + 8xy^2z)\boldsymbol{i} + (3x^3y - 3xy)\boldsymbol{j} - (4y^2z^2 + 2x^3z)\boldsymbol{k}$ is solenoidal but $B = xyz^2\boldsymbol{A}$ is solenoidal.
2. Show that $\boldsymbol{A} = (6xy + z^3)\boldsymbol{i} + (3x^2 - z)\boldsymbol{j} - (3xz^2 - y)\boldsymbol{k}$ is irrotational. Find $\emptyset$ such that $\boldsymbol{A} = \boldsymbol{\nabla}\emptyset$
3. Find constants a, b, c such that $\boldsymbol{v} = (x + 2y + az)\boldsymbol{i} + (bx - 3y - z)\boldsymbol{j} + (4x + cy + 2z)\boldsymbol{k}$ is irrotational. Show that $\boldsymbol{v}$ can be expressed as the gradient of a scalar function.
4. A fluid motion is given by $\boldsymbol{v} = (y + z)\boldsymbol{i} + (z + x)\boldsymbol{j} + (x + y)\boldsymbol{k}$. Show that the motion is irrotational and hence find the velocity potential.

Assignment:

Topic:

Week-6: Vector Integration

Learning Outcomes:

Able to analyze the conceptual and implementation aspects of vector integration, surface integration, volume integration

W-6	C-16	Vector integration, line integration, problem-solving	CO1	TLS1
	C-17	Surface integration, problem-solving	CO3	
	C-18	Volume integration, problem-solving	CO3	

Reference:

Vector Analysis by Murray R. Spiegel, Ph. D, Chapter 5, Vector Integration
Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors

Applications:

- A line integral is used to calculate the mass of wire.
- It helps to calculate the moment of inertia and centre of mass of wire.
- It is used in Ampere's Law to compute the magnetic field around a conductor.
- In Faraday's Law of Magnetic Induction, a line integral helps to determine the voltage generated in a loop.
- Line integral helps to calculate the work done by a force on a moving object in a vector field.

Review Question:

1. Find the total work done in moving a particle in a force field given by $\boldsymbol{F} = 3xy\boldsymbol{i} - 5z\boldsymbol{j} + 10x\boldsymbol{k}$ along the curve $x = t^2 + 1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to $t = 2$.
2. If $\boldsymbol{A} = 4xz\boldsymbol{i} - y^2\boldsymbol{j} + yz\boldsymbol{k}$, evaluate $\iint_S \boldsymbol{F}.\boldsymbol{n}dS$, where S is the surface of the cube bounded by $x = 0,\ x = 1,\ y = 0,\ y = 1,\ z = 0,\ z = 1$.
3. If $\boldsymbol{A} = 2xz\boldsymbol{i} - x\boldsymbol{j} + y^2\boldsymbol{k}$, evaluate $\iiint_V \boldsymbol{F}dV$, where V is the region bounded by the surfaces $x = 0,\ y = 0,\ y = 6,\ z = x^2,\ z = 4$.
4. The acceleration a of a particle at any time $t \geq 0$ is given by $\boldsymbol{a} = e^{-t}\boldsymbol{i} - 6(t+1)\boldsymbol{j} + 3sint\boldsymbol{k}$. If the velocity v and displacement r are zero at $t = 0$, find v and r at any time.

Assignment:

Topic:

Class Test-02

Week & Class No.		Topic to be covered	CO	Teaching & Learning Strategies (TLS)
Week-7: Divergence theorem, Stokes theorem and related theorem **Learning Outcomes:** Able to analyze the conceptual and implementation aspects of Divergence Theorem, Stokes Theorem, Green's Theorem				
W-7	C-19	Green's Theorem, statement, proof, problem-solving	CO1	TLS1
	C-20	Divergence Theorem, Stokes Theorem, statement, problem-solving	CO3	TLS3
	C-21	Class test from previous classes		

Reference:

Vector Analysis by Murray R. Spiegel, Ph. D, Chapter 6 , Divergence Theorem, Stokes Theorem, Green's Theorem

Advanced Engineering Mathematics by H K Dass, Chapter 5, Vectors

Applications:
• **Green's Theorem:** In computer graphics, Green's Theorem can be applied to compute areas and centroids of 2D shapes, which is useful in image processing and computer-aided design. • **Divergence Theorem:** In machine learning, the divergence theorem can be used to understand the flow of information through a system. It has applications in feature selection, where the divergence of information measures the importance of features in a dataset. • **Stokes' Theorem:** In robotics, Stokes' Theorem can be applied to compute the rotation of a robot's end-effector path, relating the circulation of the velocity vector around a closed loop to the flux of the vorticity vector through a surface enclosed by the loop.
Review Question:
1. Verify Green's theorem in the plane for $\oint_C (xy + y^2)dx + x^2 dy$ where C is the closed curve of the region bounded by $y = x$ and $y = x^2$. 2. Let $\phi = 45x^2 y$ and let V denote the closed region bounded by the planes $4x + 2y + z = 8,\ x = 0,\ y = 0,\ z = 0$ evaluate $\iiint_V \phi dV$.
Assignment:
Topic:

Mid-term Break

Week & Class No.		**Topic to be covered**	**CO**	**Teaching & Learning Strategies (TLS)**
Week-8: Matrices **Learning Outcomes:** Able to understand the matrix, matrix algebra, multiplication of matrix, inverse of the matrix				
W-8	C-22	Introduction, matrix, definition, types of matrices, examples. Adjoint matrix, the inverse of a matrix	CO1	
	C-23	Matrix polynomials, eigen-values	CO2	
	C-24	Eigenvectors, problem solving	CO2	

Reference:
Matrices by P. N. Chatterjee, Chapter 1, Matrices

Applications:

- Matrices, being the organization of data into columns and rows, can have many applications in representing demographic data, in computer and scientific applications, among others**.** They can be used as a representation of data or as a tool to solve specific problems.
- Matrices are used to represent real-world data about specific populations, such as the number of people who have a specific trait. They can also be used to model projections in population growth.
- In the field of computing, matrices are used in message encryption. They are used to create three-dimensional graphic images and realistic looking motion on a two-dimensional computer screen and also in the calculation of algorithms that create Google page rankings.
- Matrices are used to compress electronic information and play a role in storing fingerprint information. Errors in electronic transmissions are identified and corrected with the use of matrices.

Review Question:

1. If $A = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 7 & 8 \\ 9 & 10 & 12 \end{bmatrix}$ find inverse of the matrix A.
2. Find the eigenvalues of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$
3. Determine the characteristic equation and roots of the matrix $A = \begin{bmatrix} 1 & -1 & 4 \\ 0 & 3 & 7 \\ 0 & 0 & 5 \end{bmatrix}$
4. Find the eigenvalues of the matrix $A = \begin{bmatrix} -3 & 2 & 2 \\ -6 & 5 & 2 \\ -7 & 4 & 4 \end{bmatrix}$

Assignment:

Topic:

Week-9: Matrices

Learning Outcomes:

Able to analyze the conceptual and implementation aspects of Cayley-Hamilton theorem, rank, canonical form of the matrix

W-9	C-25	Cayley-Hamilton theorem with example and application in various fields.	CO2	
	C-26	Rank and nullity	CO2	
	C-27	Normal and canonical forms of matrix	CO3	

Applications:

- **Cryptography:** Rank properties are employed in cryptographic algorithms. Matrices with specific rank characteristics are used in the design of cryptographic systems to ensure security and resist attacks.
- **Network Analysis and Connectivity:** The rank of matrices is applied in network analysis for studying connectivity and dependencies in systems. For example, it can be used to determine the independence of paths in a computer network or to identify critical components.

Review Question:

1. Show that the matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 0 & 3 \\ 2 & -1 & 1 \end{bmatrix}$ satisfies Cayley Hamilton theorem.
2. Verify Cayley Hamilton's theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1}.
3. Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$.

Assignment:

Topic:

Week-10: Laplace Transform

Learning Outcomes:

Able to identify the fundamental of Laplace transforms and properties of Laplace transforms

W-10	C-28	Definition, Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace Transforms	CO1	TLS1
	C-29	Some important properties of Laplace transform	CO1	TLS4
	C-30	Laplace Transforms of first translation and second translation properties, Laplace Transforms of derivatives	CO2	

Reference:

- Advanced Engineering Mathematics by H. K. Dass, Chapter 13,
- Laplace Transform Mathematical Method volume two by Prof Abdur Rahman, Chapter 01

Applications:
• Laplace transform is extensively used in the analysis and design of control systems. It helps in transforming differential equations governing the behavior of mechanical systems into algebraic equations, which are easier to manipulate and analyze.
Review Question:
1. Find the Laplace transform of $3t^{-4} - 2t^3 + 4e^{-3t} - 2\sin 5t + 3\cos 2t$. **2.** Find the Laplace transform of $t \sin at$ **3.** Find the Laplace transform of $\sinh at$ and $\cosh at$ **4.** State and prove the first translation or shifting property of Laplace transform. **5.** State and prove the second translation or shifting property of Laplace transform. **6.** If $L\{F(t)\} = f(s)$ then prove that $L\{F'''(t)\} = s^3 f(s) - s^2 F(0) - sF'(0) - F''(0)$. **7.** Obtain the Laplace transform of $t^2 e^t \sin 4t$. **8.** Obtain the Laplace transform of $\frac{\sin 2t}{t}$. **9.** Prove that $\int_0^\infty te^{-st} \sin at\, dt = \frac{2as}{(s^2+a^2)^2}$
Assignment:
Topic:
Class Test-3

Week & Class No.		**Topic to be covered**	**CO**	**Teaching & Learning Strategies (TLS)**
Week-11: Laplace Transform **Learning Outcomes:** Able to identify the fundamental properties of the unit step function, Periodic function				
W-11	C-31	The unit step function, Periodic function	CO2	TLS1
	C-32	Problem solving from periodic function	CO2	
	C-33	Inverse Laplace Transforms	CO2	
Reference: • Advanced Engineering Mathematics by H. K. Dass, Chapter 13, • Laplace Transform Mathematical Method volume two by Prof Abdur Rahman, Chapter 01				
Applications:				
• Laplace transform is used to study the response of mechanical systems to different types of inputs, such as step inputs, impulse inputs, or sinusoidal inputs. This analysis is crucial for understanding how a system behaves under different conditions.				

- Laplace transform is applied to analyze stress and strain distributions in mechanical components subjected to dynamic loads. This is particularly useful in fatigue analysis and the study of material behavior under varying conditions.

Review Question:

1. Express the following function in terms of unit step functions and find its Laplace transform:

$$f(t) = \begin{cases} 8, & t < 2 \\ 6, & t > 2 \end{cases}$$

2. Find the Laplace transform of $F(t)$ when $F(t)$ is a periodic function with period 2π such that $F(t) = \begin{cases} \sin t\,, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \end{cases}$
3. Find the Laplace transform of the waveform $f(t) = \left(\frac{2t}{3}\right), 0 \le t \le 3$.

Assignment:

Topic:

Week-12: Inverse Laplace Transform

Learning Outcomes:

Able to understand the Convolution theorem

W-12	C-34	Properties of Inverse Laplace Transforms	CO2	TLS2
	C-35	Problem solving from Inverse Laplace Transforms	CO2	TLS4
	C-36	Convolution theorem	CO3	

Reference:

- Advanced Engineering Mathematics by H. K. Dass, Chapter 13,
- Laplace Transform Mathematical Method volume two by Prof Abdur Rahman, Chapter 02

Applications:

- Mechanical systems can experience shocks and vibrations due to various factors like impact, earthquakes, or machinery operation. The inverse Laplace transform helps in analyzing the time-domain response of mechanical systems to these dynamic inputs. Engineers can use this information to design structures and components that can withstand or mitigate the effects of shocks and vibrations.

Review Question:

1. Evaluate $L^{-1}\left\{\frac{5s-6}{s^2+9} - \frac{s-15}{s^2-25}\right\}$

2. Evaluate $L^{-1}\left\{\dfrac{6s-4}{s^2-4s+20}\right\}$

3. Evaluate the followings using the convolution theorem:

(i) $L^{-1}\left\{\frac{1}{(s+1)(s^2+1)}\right\}$ (ii) $L^{-1}\left\{\frac{1}{s^2(s^2+3)}\right\}$

4. Evaluate $L^{-1}\left\{\dfrac{1}{s^2(s+1)^2}\right\}$ by using the convolution theorem.

Assignment:

Topic:

Week-13: Inverse Laplace Transform

Learning Outcomes:

Able to identify the fundamental properties of partial fractions

W-13	C-37	Problem solved by using convolution theorem	CO3	TLS1
	C-38	Partial fractions	CO3	TLS4
	C-39	Problem solved by using partial fractions	CO3	

Reference:

- Advanced Engineering Mathematics by H. K. Dass, Chapter 13,
- Laplace Transform Mathematical Method volume two by Prof Abdur Rahman, Chapter 02

Applications:

- Laplace transforms is commonly employed in the development of mathematical models for mechanical systems. The inverse Laplace transform allows engineers to simulate and analyze the time-domain behavior of these models, aiding in the prediction of system responses under different conditions.

Review Question:

- Prove that $L^{-1}\left\{\dfrac{1}{s^{3/2}}\right\}=2\sqrt{\dfrac{t}{\pi}}$
- Prove that $L^{-1}\left\{\frac{1}{s^3(s^2+1)}\right\}=\frac{t^2}{2}+cost-1.$
- Evaluate $L^{-1}\left\{\dfrac{5s+3}{(s-1)(s^2+2s+5)}\right\}$ using partial fraction.

Assignment:

Topic:

Class Test-4

Week & Class No.		Topic to be covered	CO	Teaching & Learning Strategies (TLS)
Week-14: Inverse Laplace Transform **Learning Outcomes:** Able to identify the fundamental properties of Solutions of differential equations by Laplace Transforms				
W-14	C-40	Solutions of differential equations by Laplace Transforms	CO3	
	C-41	ODE and PDE by Laplace Transforms	CO3	
	C-42	Review class		

Reference:

- Advanced Engineering Mathematics by H. K. Dass, Chapter 13,
- Laplace Transform Mathematical Method volume two by Prof Abdur Rahman, Chapter 02

Applications:

- In the study of transient heat conduction and thermal systems, Laplace transforms are used to analyze temperature distributions over time. The inverse Laplace transform helps engineers obtain solutions in the time domain, allowing them to understand how temperature varies with time in different components and materials.

Review Question:

1. Solve the differential equation by Laplace transform: $y'' + y' - 2y = sint;\ y(0) = 0,\ y'(0) = 0.$
2. Solve the differential equation by Laplace transform:
   $y'' - 3y' + 2y = 4t + e^{3t};\ \ y(0) = 1,\ y'(0) = -1.$
3. Find using Laplace transform the solution of the following differential equation which satisfy the given conditions: $y'' - 9y = e^{t};\ \ y(0) = 1,\ y'(0) = 0$. Also verify the result.

Assignment:

Topic:

Prepared by:	Checked by:	Approved by: (Head of the Dept.)
Md. Abdullah Al Mamun **Assistant Professor & Head** **Dept. of Mathematics** **BAUET**		